

北北基高級中等學校

113 學年度學科能力測驗聯合模擬考試

英文考科

請於考試開始鈴響起，在答題卷簽名欄位以正楷簽全名

—作答注意事項—

考試範圍：第 1～3 冊

考試時間：100 分鐘

作答方式：

- 選擇題用 2B 鉛筆在「答題卷」上作答；更正時以橡皮擦擦拭，切勿使用修正帶（液）。
- 除題目另有規定外，非選擇題用筆尖較粗之黑色墨水的筆在「答題卷」上作答；更正時，可以使用修正帶（液）。
- 考生須依上述規定劃記或作答，若未依規定而導致答案難以辨識或評閱時，恐將影響成績。
- 答題卷每人一張，不得要求增補。

選擇題計分方式：

- 單選題：每題有 n 個選項，其中只有一個是正確或最適當的選項。各題答對者，得該題的分數；答錯、未作答或劃記多於一個選項者，該題以零分計算。
- 多選題：每題有 n 個選項，其中至少有一個是正確的選項。各題之選項獨立判定，所有選項均答對者，得該題全部的分數；答錯 k 個選項者，得該題 $\frac{n-2k}{n}$ 的分數；但得分低於零分或所有選項均未作答者，該題以零分

祝考試順利



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第壹部分、選擇題（占 62 分）

一、詞彙題（占 10 分）

說明：第 1. 題至第 10. 題為單選題，每題 1 分。

1. After studying diligently for weeks, Emily felt confident that she had acquired _____ knowledge to pass the exam successfully.
(A) countless (B) efficient (C) sufficient (D) precautions
2. Jacob submitted his job _____ online and eagerly awaited a response, hoping to hear back from the company soon.
(A) education (B) relation (C) subscription (D) application
3. The company has to _____ unnecessary expenses if it wishes to increase profits.
(A) refresh (B) treat (C) ignore (D) eliminate
4. After the surgery, the patient's condition worsened _____, prompting the medical team to reassess and adjust the treatment urgently.
(A) unexpectedly (B) conventionally (C) effectively (D) previously
5. The _____ between maintaining tradition and pursuing economic growth has been a heatedly debated issue in many cultural discussions.
(A) addict (B) conflict (C) influence (D) function
6. Health officials are urging people to get vaccinated as soon as possible to reduce the risk of being _____ with the seasonal flu.
(A) affected (B) infected (C) detected (D) selected
7. The Lin family enjoyed spending time together doing _____ chores on weekends like cleaning, cooking, and gardening.
(A) physical (B) domestic (C) durable (D) genuine
8. Stella knows how important it is to stay connected, so she always keeps her phone charger _____ in her bag in case the battery runs low.
(A) handy (B) suitable (C) remarkable (D) apparent
9. Out of love, parents sometimes go to _____ to protect their children, doing everything they can to keep them safe and secure.
(A) contacts (B) cells (C) extremes (D) yards
10. After several famous Korean idols chose to _____ suicide, there has been an increased awareness of the need for mental health support within the K-pop community.
(A) organize (B) judge (C) direct (D) commit

二、綜合測驗（占 10 分）

說明：第 11.題至第 20.題為單選題，每題 1 分。

第 11. 至 15. 題為題組

Pollinators are disappearing, which is a big problem. It affects not only our ecosystems but also our well-being. Bees, butterflies, wasps, beetles, bats, flies, and hummingbirds are all important pollinators. Unfortunately, these pollinators are decreasing globally because of 11., pesticide use, and climate change. The possibility of a global food shortage due to the extinction of bees is worrying. Unfortunately, Robot bees offer a potential solution to this problem. Researchers and engineers around the world have been trying out the ways 12. robots can perform the tasks of real bees. Robot bees are machines designed to imitate the work of real bees. They are mainly used for pollination but can also monitor hive health.

Perhaps the biggest challenge for robot bees is to perform the tasks of real bees. Real bees can fly for hours and remain stable in wind and rain. They can find flowers that are miles away from their hives and use advanced senses of smell and sight to locate the right flowers and the right parts of those flowers to collect and 13. pollen. These abilities and behaviors come naturally to real bees after millions of years of 14.. However, they are extremely difficult for even the smartest robots to replicate. Despite this challenge, we may not have much choice. The global population is expected to increase by 25 percent to about 10 billion people by 2050. To meet the demand, food production will need to increase significantly, as will the number of pollinators. We may need the assistance of robot bees to help us 15. these challenges.

- | | | | |
|--------------------------------|-------------------------|----------------|----------------|
| 11. (A) automatic transmission | (B) habitat destruction | | |
| (C) further investigation | (D) general consensus | | |
| 12. (A) which | (B) what | (C) for which | (D) in which |
| 13. (A) prevent | (B) represent | (C) deposit | (D) subtract |
| 14. (A) permission | (B) evolution | (C) separation | (D) definition |
| 15. (A) resort to | (B) call for | (C) show off | (D) cope with |

第 16. 至 20. 題為題組

Studying effectively is essential for academic success. Developing good study habits can help students excel and achieve their academic goals. One key aspect of effective studying is time management. By creating and 16. a study schedule, students can allocate enough time to each subject. Besides managing time, students should also focus on critical comprehension. Merely memorizing information may not be enough for long-term success. Instead, they should aim to understand the underlying concepts and principles. This deeper understanding will make 17. easier to apply knowledge in different contexts and improve retention.

Another important aspect of effective studying is seeking clarification when 18. difficulties. Students should not hesitate to ask teachers or peers for help. Discussing challenging topics with others can facilitate learning and lead to a better grasp of the subject matter. Additionally, using a variety of

study methods can enhance learning. Techniques such as summarizing information, making mind maps, and practicing past exam questions can be very beneficial. These methods help reinforce understanding and identify any gaps in knowledge.

Moreover, maintaining a balanced lifestyle is crucial. Students should 19. they get enough rest, exercise, and leisure activities. A healthy mind and body can significantly improve concentration and productivity. 20., effective studying involves good time management, critical comprehension, seeking help when needed, using diverse study techniques, and maintaining a balanced lifestyle. By adopting these practices, students can improve their academic performance and achieve their goals.

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|---------------------|-----------------|------------------|------------------|
| 16. (A) adapting to | (B) sticking to | (C) applying to | (D) referring to |
| 17. (A) it | (B) this | (C) so | (D) that |
| 18. (A) generating | (B) launching | (C) deliberating | (D) encountering |
| 19. (A) blend | (B) acquire | (C) ensure | (D) spray |
| 20. (A) By contrast | (B) In summary | (C) Until now | (D) For example |

三、文意選填（占 10 分）

說明：第21.題至第30.題為單選題，每題 1 分。

第 21. 至 30. 題為題組

Have you ever been bombarded with various criticisms from different people but chosen to ignore them and insist that you are right? In this case, you may have experienced the Dunning-Kruger effect. The concept of the Dunning-Kruger effect 21. in a 1999 paper by Cornell University psychologists David Dunning and Justin Kruger. The participants took a test assessing their logic, grammar, and sense of humor, and the results revealed that those who performed poorly 22. their skills as far above average.

The Dunning-Kruger effect is a(n) 23. bias in which individuals tend to overestimate their knowledge or ability in a specific area. Two main 24. can explain this. Firstly, lacking metacognition, people describe themselves as above average in intelligence, humor, and other skills, which restrains them from 25. their real capabilities and learning progress. Thus, they can't acknowledge their incompetence. Secondly, this 26. shows that some people lack the self-awareness to evaluate their skills. As a(n) 27., after they acquire a small amount of knowledge in a particular field, the phenomenon occurs. They are satisfied with the content they've obtained, which 28. inappropriate confidence. This inflated confidence makes them feel like experts suddenly.

To avoid this effect, people should make a constant effort to learn about a new topic and challenge themselves to consider that their 29. might be wrong. Only after continuing to explore a new subject do they realize how extensive it is and how far they should go before mastering it. In addition, people ought to seek 30. from peers while learning so as not to become overconfident. Following these tips can help people avoid falling into the trap of the Dunning-Kruger effect.

- | | | | | |
|------------|-----------------|----------------|--------------|-----------------|
| (A) rated | (B) cognitive | (C) originated | (D) enhances | (E) tendency |
| (F) causes | (G) consequence | (H) perceiving | (I) feedback | (J) assumptions |

四、篇章結構（占 8 分）

說明：第31.題至第34.題為單選題，每題 2 分。

第 31. 至 34. 題為題組

Since the year of 1927, *TIME* magazine has annually selected a Person of the Year, who represents the individual most influencing the headlines over the past year. Traditionally, this honor has often been awarded to male figures, mostly politicians or business magnates. 31.

However, in 2023, the person that emerged as the most distinguished and the brightest star defied these conventional roles entirely. Taylor Swift was everywhere in 2023, breaking records and filling stadiums. She reached a never-before-seen level of impact, blending artistry and commerce in an incomparable manner. She has obtained matchless achievements, including notching more No. 1 albums than any other female artist in history. 32. Particularly significant was Swift's ability to create a space for thousands and thousands of people to rediscover the communal experiences that were taken from us during the pandemic, offering joy to the fans each night at her monumentally successful Eras Tour. Swift transcended national borders to become a ray of positivity for countless individuals. Her ability to connect with people on a global scale surpasses that of any other contemporary figure. 33. Literal effect caused by her fans, who call themselves Swifties, was shown on data everywhere.

Moreover, Swift's impact extended beyond mere statistics. She bravely embraced her own vulnerabilities, inspiring others to do the same and fostering a sense of self-awareness among her audience. Through her own journey, she crafted a narrative that resonated with millions and provided them with not just entertainment but also a mirror for self-reflection.

34. For her remarkable ability to create a world that embraced diversity and brought happiness to a society in need, Taylor Swift rightfully earned the title of *TIME*'s 2023 Person of the Year.

- (A) Her influence was evident worldwide.
- (B) Numerous U.S. Presidents, leaders of Russia or Popes have been receivers of this recognition.
- (C) Through entertaining and songwriting, this talented artist did what she loves and does better than others.
- (D) Some may attribute Swift's success to the planets and the fate, but that just overlooks her good ideas and power moves.

五、閱讀測驗（占 24 分）

說明：第35.題至第46.題為單選題，每題 2 分。

第 35. 至 38. 題為題組

Law enforcement authorities have achieved a notable breakthrough in destroying LockBit, identified by Europol as the "world's largest ransomware operation." Led by the National Crime Agency (NCA) of the United Kingdom, a union of international agencies collaborated in "Operation Cronos," aimed at disrupting the activities of the ransomware-as-a-service gang. The NCA's infiltration of LockBit's network resulted in the arrest of two suspects and the seizure of over 200 cryptocurrency accounts.

Operation Cronos, involving agencies from various countries such as the United States, France, Germany, and others, led to the closure of 28 servers and LockBit's public leak site. Additionally, the NCA acquired LockBit's source code, decryption keys, and crucial intelligence on the gang's operations.

Moreover, the operation unveiled that ransomware gangs often maintain victims' data even after receiving ransom payments, indicating the criminals' untrustworthiness. Data belonging to paying victims was discovered on LockBit's systems, highlighting the lack of assurance from criminals. Europol disclosed the identification of over 14,000 "rogue accounts" used by LockBit and gained LockBit's custom data leakage tool, known as StealBit.

This collaborative international effort stresses the significance of cooperation in combating cybercrime. The dismantling of LockBit represents a significant milestone in the fight against ransomware, emphasizing the importance of ongoing collaboration among law enforcement agencies worldwide.

35. What was the primary objective of "Operation Cronos" as mentioned in the article?

- (A) To enhance cybersecurity measures worldwide.
- (B) To destroy the LockBit ransomware group.
- (C) To promote international cooperation among law enforcement agencies.
- (D) To investigate cybercrimes in various countries.

36. What did the operation reveal about the behavior of ransomware gangs, as mentioned in the passage?

- (A) After receiving ransom payments, they instantly return data to the victims.
- (B) They frequently cooperate with law enforcement agencies to resolve cases.
- (C) They retain victims' data even after ransom payments have been received.
- (D) They rarely possess custom data leakage tools.

37. What significant assets were tracked down during the operation against LockBit?

- (A) Over 28 Bitcoin wallets.
- (B) Their source code and decryption keys.
- (C) 200 servers hosted by ransomware operations.
- (D) Bank account details of victims.

38. According to the article, what is emphasized as crucial in fighting cybercrime?

- (A) Developing advanced encryption methods.
- (B) International cooperation among law enforcement agencies.
- (C) Expanding monitoring on dark web activities.
- (D) Providing ransom payments to hackers.

第 39. 至 42. 題為題組

A QR Code, a quick-response code, is a special kind of barcode made up of a square pattern with black and white squares. QR Code was designed by a Japanese engineer, Masahiro Hara. After completing studies at Hosei University, Hara joined Denso, a subsidiary of the Toyota Group. In 1992, he was assigned to create a new 2D code for tracking automotive components efficiently. Inspired by a game of go during lunch break, Hara realized that its black and white pattern could encode data. He thus conducted research to design optimal position patterns for readability of the code. Finally, he successfully created QR code in 1994, which revolutionized tracking systems.

The uses of QR codes vary across different contexts. They provide virtual event access, offer multimedia content, and facilitate mobile shopping experiences. As for shopping, QR codes streamline payments, store bank details securely, and support various digital payment platforms. Take restaurants for example. QR codes make it possible for consumers to view menus, place orders, and make payments without lining up or interacting with staff, which ensures safety during the COVID-19 pandemic, and thus enhances convenience in dining experiences. Other industries also benefit from QR codes. Obviously, the widespread adoption **underscores** QR codes' role in modernizing retail and hospitality industries.

However, QR codes are not without shortcomings. First, not every smartphone can scan QR codes directly, which sometimes requires users to download additional apps. Second, they rely on the Internet connectivity. Users who are denied Internet access may fail to use them. Third, there is always the risk of personal data being leaked through scanning a seemingly legitimate QR code. Last, QR codes provide one-way communication only, hindering effective interactions, particularly in customer support settings. Despite the drawbacks mentioned above, it is believed that QR code technology will evolve, and its role in advancing technology industry is expected to persist.

39. What is the purpose of this passage?

- (A) To illustrate the history of QR codes since the industrial revolution.
- (B) To explain the way QR codes are generated and their structure.
- (C) To compare and contrast different types of QR codes.
- (D) To present an overall introduction to QR codes.

40. What does the word “**underscore**” mean in the second paragraph?

- (A) To produce.
- (B) To disappear.
- (C) To emphasize.
- (D) To deliver.

41. According to the passage, which of the following statements is true?

- (A) Scanning QR codes may violate users' privacy and reveal personal information.
- (B) Restaurant customers using QR codes may speed up the spread of disease.
- (C) Masahiro Hara got his ideas of encoding QR codes from the chess.
- (D) People scanning QR codes may get immediate interaction or feedback.

42. Which of the following proverbs and quotes best describes Masahiro Hara's life experience, according to the passage?

- (A) People who live in glass houses shouldn't throw stones at others.
- (B) Those who can imagine anything, can create the impossible.
- (C) Intelligence plus character — that is the goal of real education.
- (D) Living without an aim is like sailing without a compass.

第 43. 至 46. 題為題組

Travel is a right that everyone should be able to enjoy, whether they are disabled or not. To allow everyone to experience our amazing world, we need to make travel accessible to all. While some disabilities are noticeable, others are not. Airports in North America and Europe are now taking steps to help travelers with invisible disabilities feel less stressed and anxious.

A new program in these airports aims to assist travelers with invisible disabilities like dementia, Alzheimer's, brain injuries, autism, and mild cognitive impairment. As part of this program, sunflower lanyards are given to these travelers to identify their memory or sensory needs. Some airports also provide quiet rooms, noise-cancelling headphones, and support from airline staff.

This initiative began at London's Gatwick Airport in 2016, where discreet lanyards were introduced for passengers with invisible disabilities to show they might need extra help or time while traveling. The Hidden Disabilities Sunflower lanyard program started by Gatwick Airport is now used in over 200 airports worldwide.

The aim of the Sunflower lanyard is to give people with hidden disabilities the confidence to travel independently, knowing they will get the support they need at every part of their journey — from the terminal to boarding, during the flight, and upon arrival. There is a false belief that people with disabilities do not want to or cannot travel. In reality, they do and they can. The lanyards inform airport staff that these travelers might need more attention, patience, or help.

43. What is this passage mainly about?

- (A) Airports take a new step to reduce pressure on people with invisible disabilities.
- (B) Sunflower lanyards aren't just for travelers at the airports but for people in daily life.
- (C) Travel is a right that should be accessible by everyone who would like a trip.
- (D) Staff and crew in the airports treat the passengers with extra patience and help.

44. According to the passage, which of the following statements is true?

- (A) The Sunflower lanyard is a new method to help people with hidden disabilities can travel alone.
- (B) In European and North American airports, a new policy helps tourists with invisible disabilities to release pressure.
- (C) Thousands of businesses across the globe have joined the global Sunflower network.
- (D) The group of experts in dementia and Alzheimer's helped use lanyards in educational institutions.

45. Where could this passage most likely be found?

- (A) A travel guideline.
- (B) A healthcare magazine.
- (C) A fashion catalogue.
- (D) A medical journal.

46. Which of the following best describes the author's attitude toward sunflower lanyards?

- (A) Sarcastic.
- (B) Critical.
- (C) Positive.
- (D) Entertaining.

第貳部分、混合題（占 10 分）

說明：本部分共有 1 題組，每一子題配分標於題末。限在答題卷標示題號的作答區內作答，並依規定用筆作答。

第 47. 至 50. 題為題組

Preserving embryos and young creatures as fossils is very unusual because their tiny skeletons are easily scattered or destroyed before they can turn into fossils. However, marine animals like ichthyosaurs have a better chance of being preserved because they lived in environments where erosion was less likely. Still, successful fossilization depended on factors like slow decomposition of soft tissues, fewer scavengers, calm waters, and rapid burying process.

Chinese scientists have found fossils of ichthyosaurs dating back about 250 million years in South Majiashan, China. An ichthyosaur mother was found carrying at least three offspring. One is under the mother's body. The second is halfway out of the birth canal, making this fossil unique. The third is still inside her, waiting to be born. A more abundant fossil finding is located in dark, oily marine rocks in Holzmaden, Germany. There, thousands of well-preserved fossils dating back about 190 million years have been discovered, including ichthyosaurs, fish, and other sea creatures. What's remarkable is that many of these ichthyosaurs contain preserved embryos. These embryos, found in various layers of the rock, suggest that the area was a favored spot for ichthyosaurs to give birth over time. The embryos are quite developed, with well-formed paddles, and one is even found in the birth canal. Additionally, there are many fossils of newborns measuring between 20 and 30 inches long.

The quality of ichthyosaur fossil preservation in Holzmaden is outstanding. However, scientists are puzzled by the abundance of pregnant ichthyosaurs and young ones found there, especially considering their rarity elsewhere. This question remains intriguing, and researchers have yet to provide a definitive answer.

47. -48. 下列簡短敘述摘記上方文章重點。請從文章中找出最適當的單詞（word）填入下列句子空格中，並視句型結構需要做適當的字形變化，使句子語意完整、語法正確，並符合全文文意。每格限填一個單詞（word）。（填充，4 分）

The passage notes that marine animals, like ichthyosaurs, had a higher chance of preservation when the breakdown of their soft tissues occurred 47., and their remains were buried 48..

49. From (A) to (F) below, choose the **ONES** that are true of the fossil deposits mentioned in the passage.（多選題，4 分）

- (A) Ichthyosaurs lived in freshwater environments.
- (B) Holzmaden yields thousands of well-preserved fossils.
- (C) The embryos found are poorly developed.
- (D) Holzmaden's fossils are mainly male ichthyosaurs.
- (E) Ichthyosaurs lived in environments less prone to erosion.
- (F) Young ichthyosaurs are common in fossil sites worldwide.

50. Which noun phrase in the passage means “the channel through which a baby animal passes in order to be born naturally”?

(簡答，2 分)

第參部分、非選擇題（占 28 分）

說明：本部分共有二大題，請依各題指示作答。答案必須寫在「答題卷」標示題號之作答區內，作答時不必抄題。

一、中譯英（占 8 分）

說明：依題號將以下中文句子譯成正確、通順、達意的英文。每題 4 分，共 8 分。

1. 因應臺灣人口老化問題，政府已經提供長照服務以減少家庭的負擔。
2. 當年長者不再能獨立生活時，他們需要各式各樣的服務選項在日常生活中協助他們。

二、英文作文（占 20 分）

說明：依提示寫一篇英文作文，文長至少 120 個單詞（words）。

提示：下圖為 2024 年年初高雄「黃色小鴨」觀光熱潮的新聞圖片，請根據此圖片寫一篇英文作文。第一段描述圖片裡的內容，包含人、地、物及活動等……。第二段說明像「黃色小鴨」此類的藝術創作對城市形象及觀光旅遊的影響。



